

T-sne analysis

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T-sne mapping misunderstanding

Previous method: mapping from feature to action;

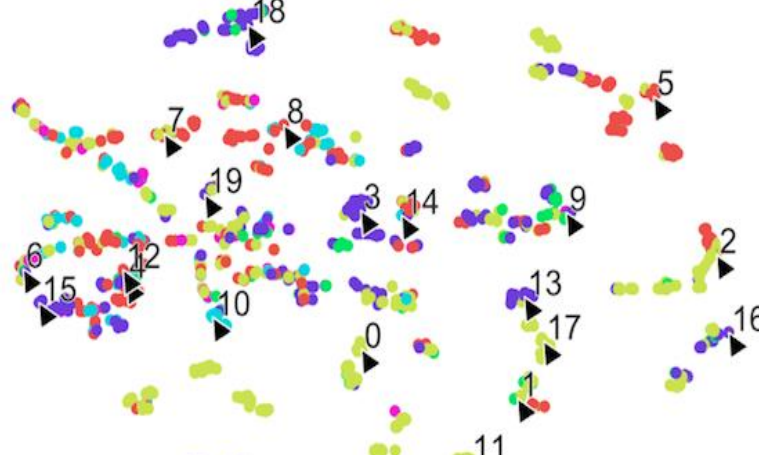
Correct method: mapping from feature to state value; by summing up the action space

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s') \sim U(D)} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i) \right)^2 \right]$$

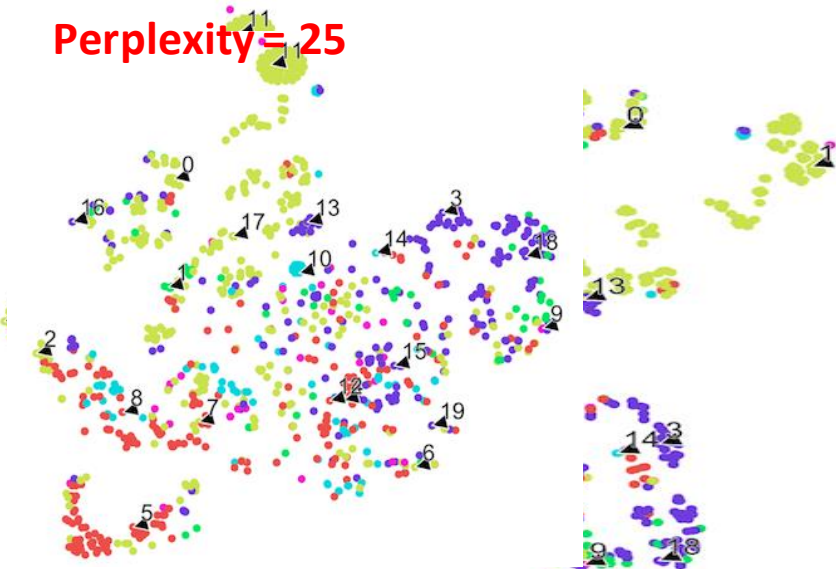
But, still show some t-sne regression work in the two following slides by mapping from feature to action.

Tsne setting regression (1)-- perplexity

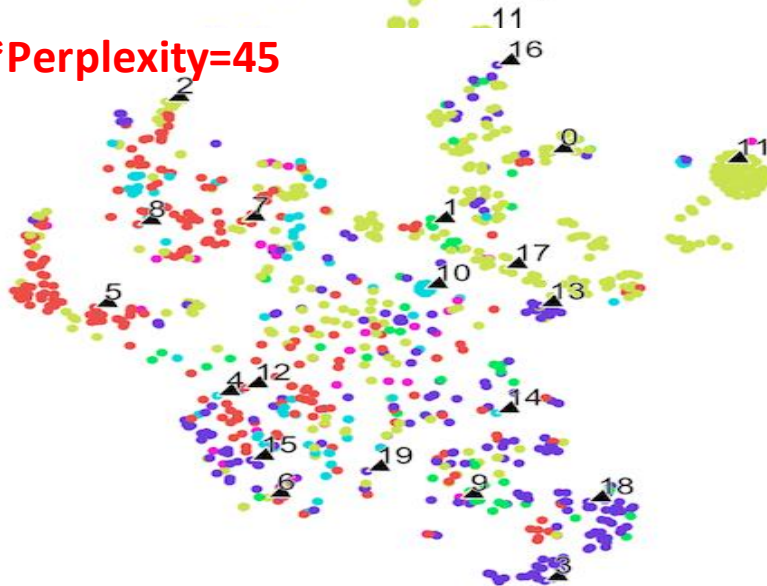
Perplexity = 5



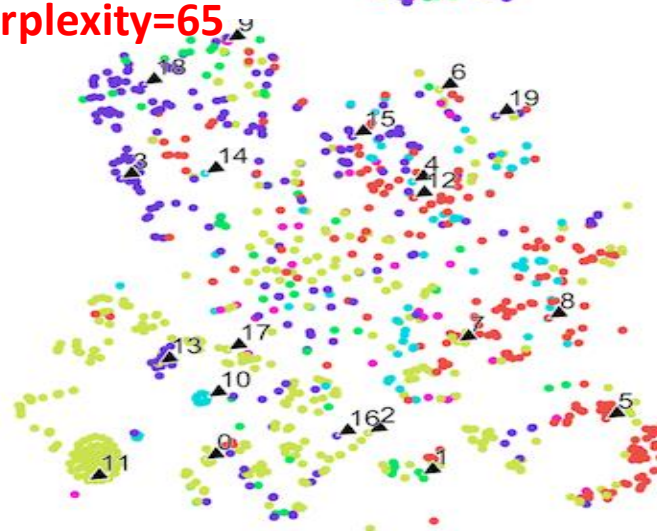
Perplexity = 25



*Perplexity = 45



Perplexity = 65



Game = "SpaceInvaders"

Data =

Dataset1(first 20k iter)

TSNE info:

Number of points = 1000

Interval = 20

Lr = 800

N_iter = 4000

Exaggeration = 12

Small Perp ==> small clusters

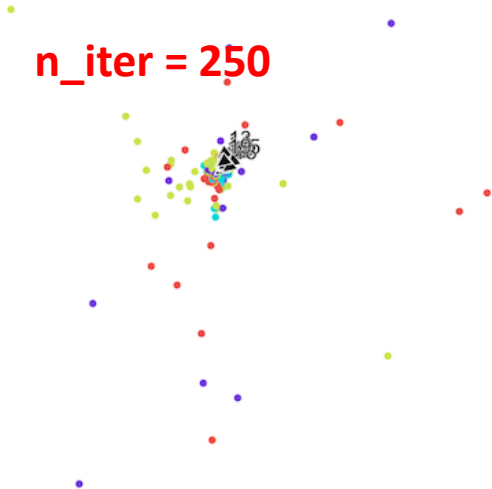
Middle perp ==> strip-like

Large Perp ==> large clusters

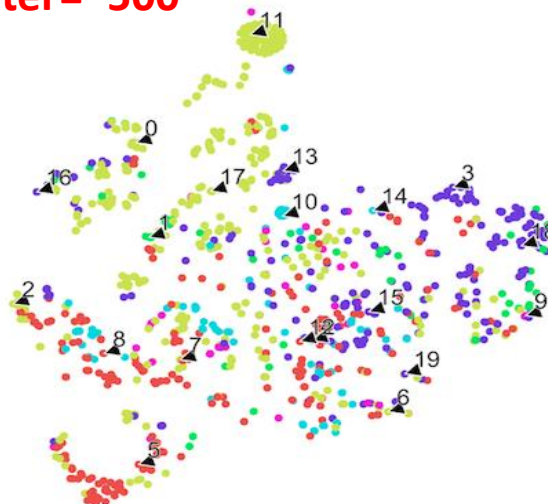
Note: numbers on figure
are the indexes of
screenshots I want to save.

Tsne setting regression (1)-- iteration

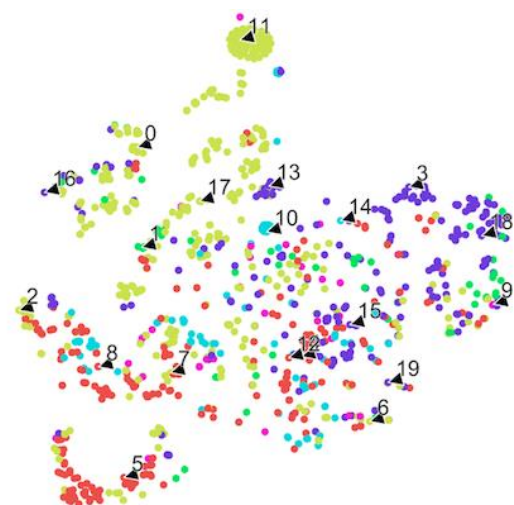
n_iter = 250



n_iter = 500



n_iter = 750



n_iter = 1000



Game = "SpaceInvaders"

Data =

Dataset1(first 20k iter)

Tsne info:

Number of points=1000

Interval=20

Perplexity=45

Lr=300

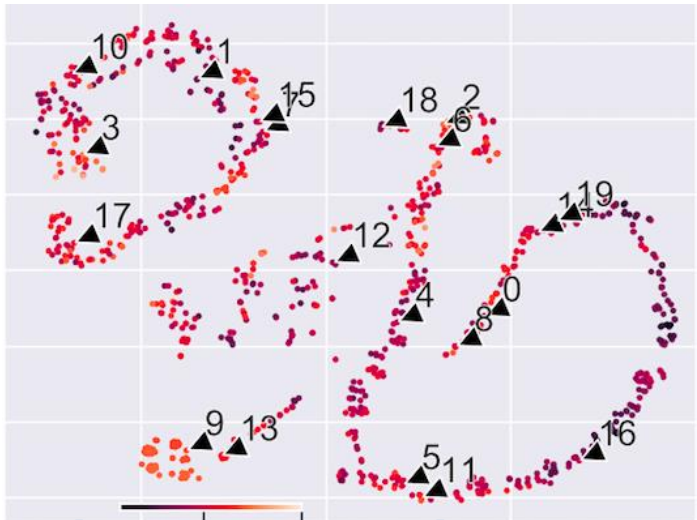
Exaggeration = 12

500 iterations is enough

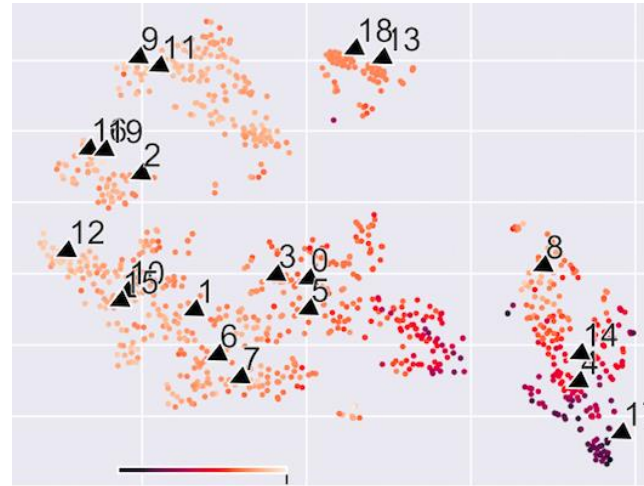
Note: optimal setting may differ from case to case, e.g. 3D tsne needs more iterations

T-sne mapping from feature to state value

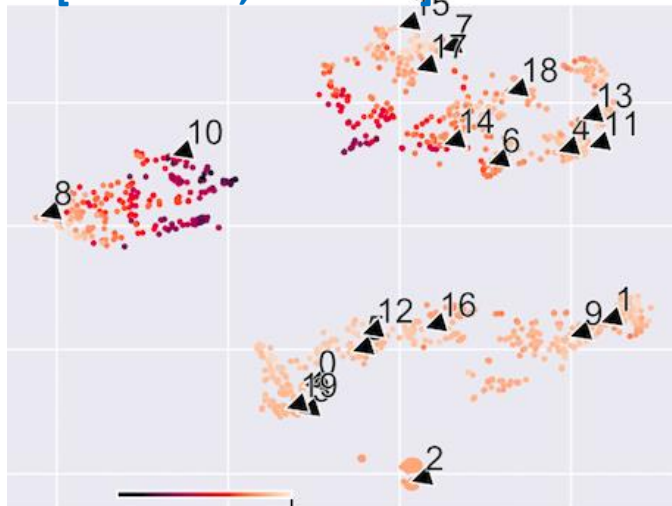
S1:Iter=[0,2000]



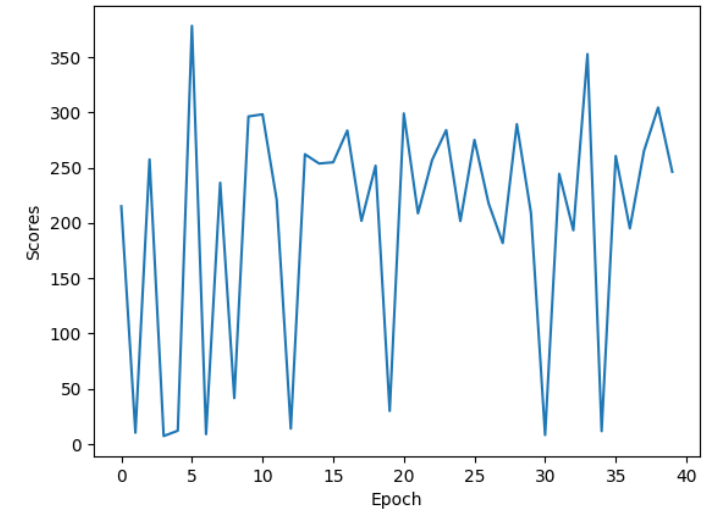
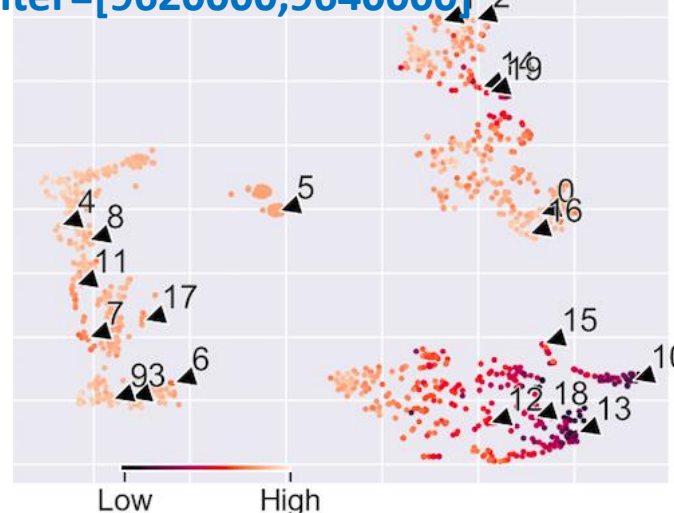
S2:Iter=[322000,3240000]



S3:Iter=[6420000,6440000]



S4:Iter=[9620000,9640000]

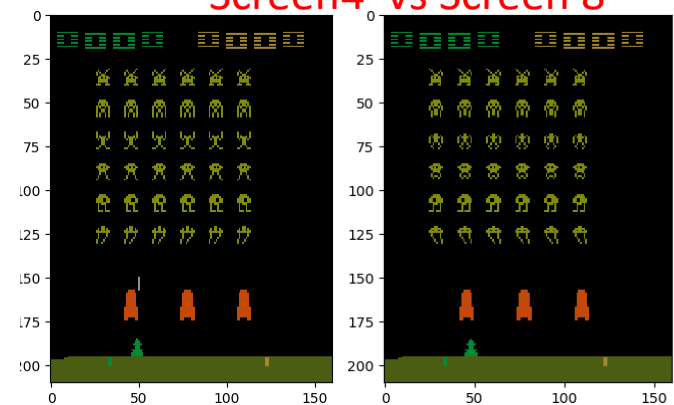


Notes: X-axis correspond to 16 million iteration, but only data before 9,640,000 are sampled and saved, Sample data every 200 iterations.

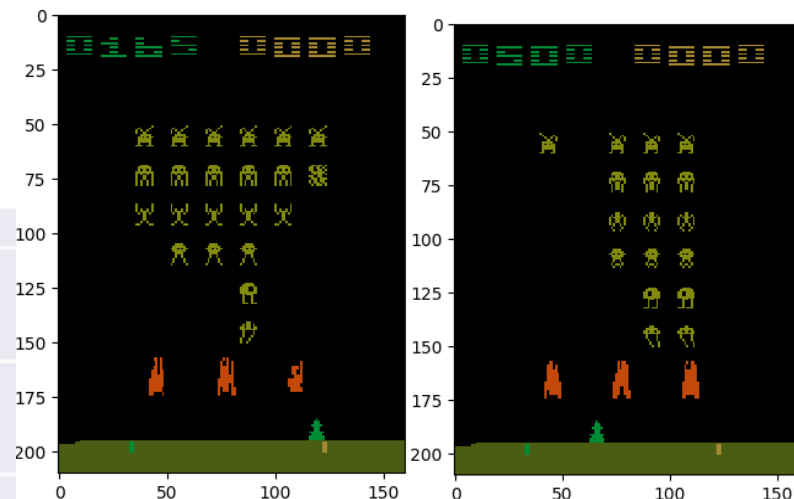
Stranges here: the score fluates dramatically, it seems that the agent does not evolve.

S4: mapping details

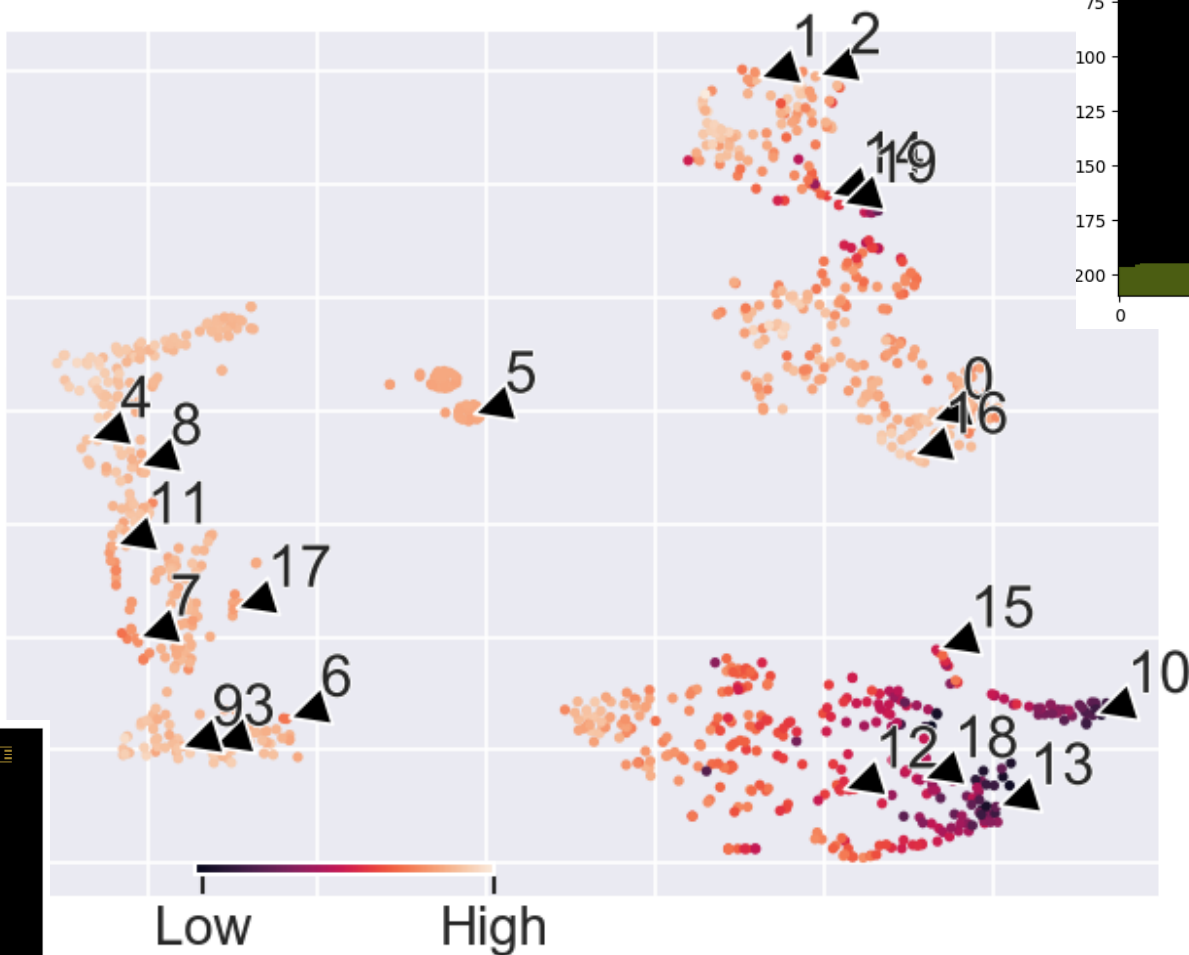
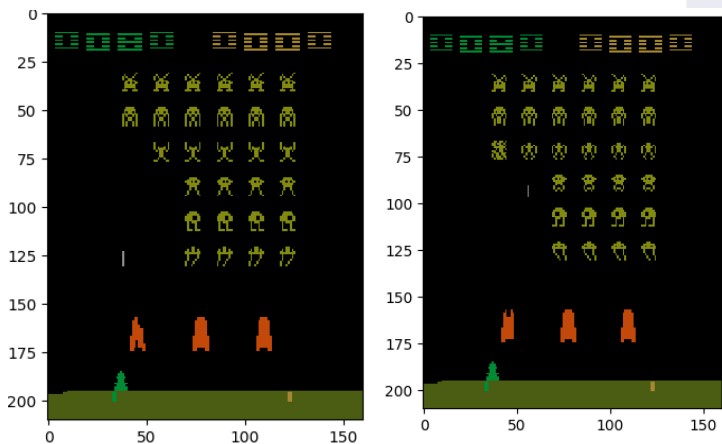
Screen4 vs Screen 8



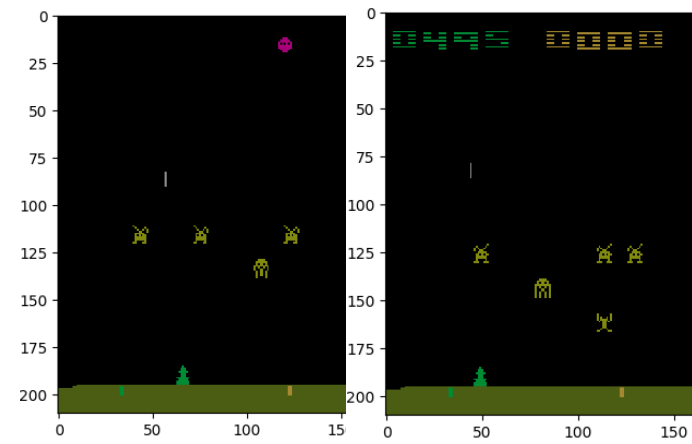
Screen14 vs Screen19



Screen3 vs Screen9

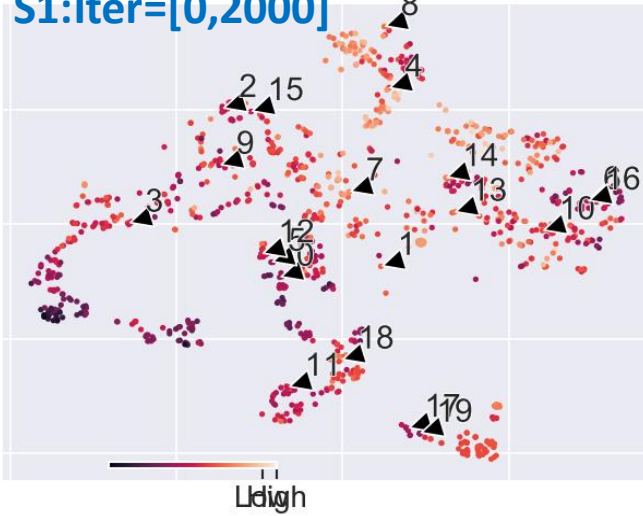


Screen10 vs Screen13

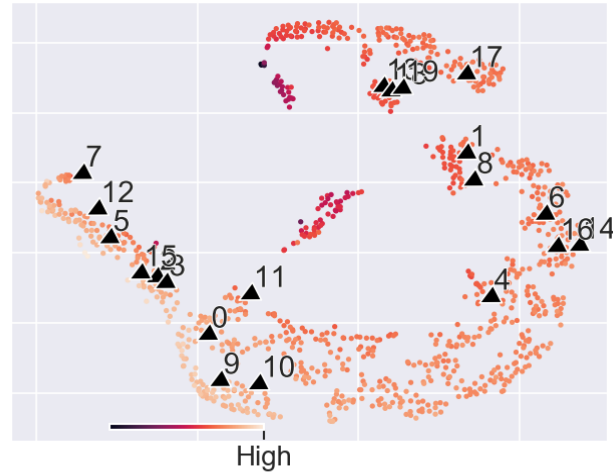


Reduced DQN hidden layer size :512=>128

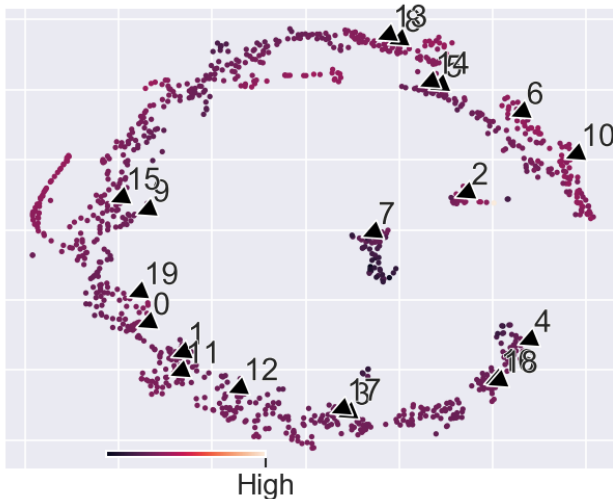
S1:Iter=[0,2000]



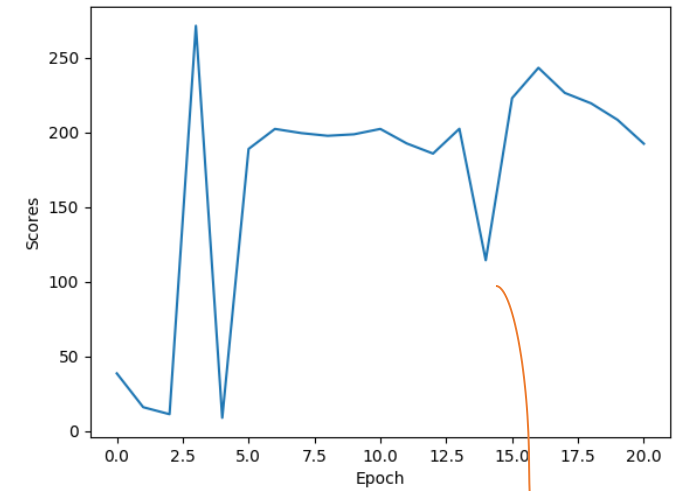
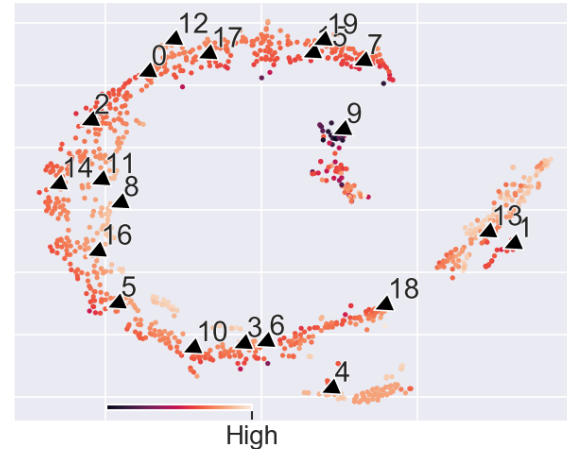
S2:Iter=[322000,324000]



S3:Iter=[6420000,6440000]



S4:Iter=[9620000,9640000]



Epoch =15 corresponds to S3
Low state value may lead to low score.